



University
of Windsor

Comp 3150 Winter 2026 (Solution)

Midterm 1

Examiner: Dr. C. I. Ezeife

Given: Thursday, Feb. 5, 2026

Student Name: _____

Student Number: _____

INSTRUCTIONS (Please Read Carefully)

Examination Period is 1 hours 20 minutes

Answer all questions. Write your answers in the spaces provided in the question paper. This is closed book and closed notes test.

Total Marks =50. Total number of sections = 2

Please read questions carefully! Misinterpreting a question intentionally or unintentionally results in getting a “ZERO” for that question. Good Luck!!!

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CONFIDENTIALITY AGREEMENT & STATEMENT OF HONESTY

I confirm that I will keep the content of this examination confidential. I confirm that I have not received any unauthorized assistance in preparing for or writing this examination. I confirm knowing that a mark of 0 may be assigned for copied work.

Student Signature

Student Name (please print)

Student I.D. Number

Date

For marking purposes only (This part not to be filled by students)

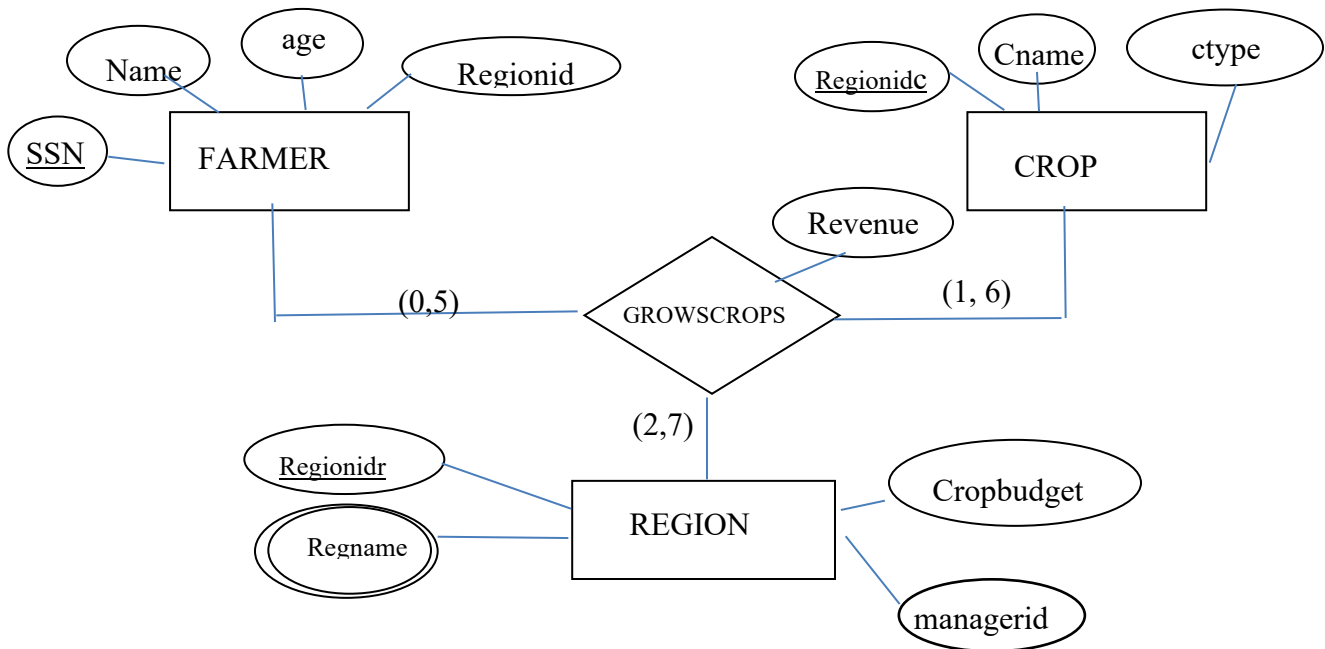
Question	Mark
Section A (15 marks for 10 multiple choice questions)	/15
Section B	
Que 1 (15 marks)	/15
Que 2 (10 marks)	/10
Que 3 (10 marks)	/10
Total	/50

Section A

15 marks for 10 Multiple Choice Questions. Each question in this section is worth 1.5 marks.

For Questions 1 – 10, circle the correct answer.

Figure A.1: The ER model Diagram for a Farmer-growscrops Database

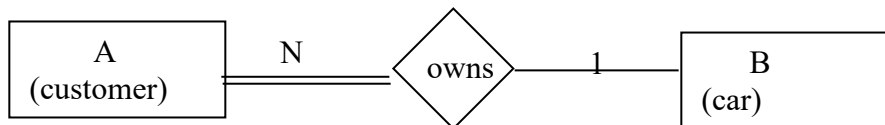


Use the ER diagram of Figure A.1 to answer the next few questions

- Which statement best describes the cardinality constraint of the relationship Growscrops shown in Figure A.1 above.
 - Each farmer can grow only one crop but each crop can be grown by many farmers.
 - Each farmer can grow many crops but each crop can be grown by only one farmer.
 - Each** farmer can grow many crops and each crop can be grown by many farmers.
 - All of the above
 - None of the above.
- Which statement best describes the participation constraint of the relationship Growscrops shown in Figure A.1 above.
 - Every farmer must grow a crop in a region

- b. **Not every farmer** has to grow a crop in a region.
 - c. Every farmer must not grow crop in a region .
 - d. All of the above.
 - e. None of the above.
- 3. Which statement best describes the entities and relationships in Figure A.1 above?
 - a. A farmer is a weak entity.
 - b. Each region can have a composite region name attribute.
 - c. **Each region** can have a multiple valued region name attribute.
 - d. All of the above.
 - e. None of the above.
- 4. What is the text relational database schema for the Relationship 'Growscrops' in Figure A.1 above?
 - a. Growscrops(Revenue).
 - b. **Growscrops(SSN, Regionidc, Regionidr, Revenue).**
 - c. Growscrops(SSN, Regionidc, Revenue).
 - d. All of the above.
 - e. None of the above.

Figure A.2 :



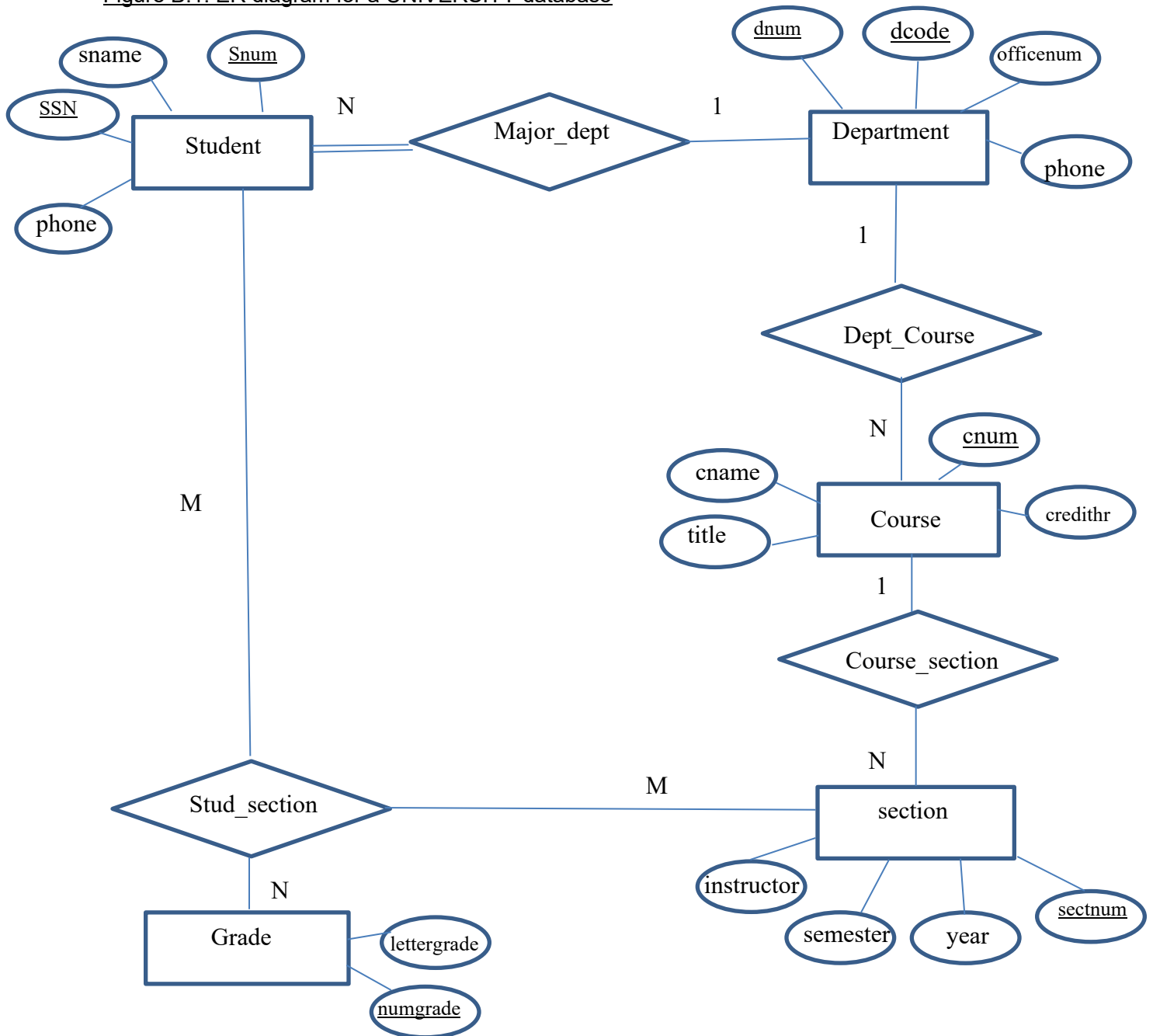
- 5. From the database on ER diagram of Figure A.2, it can be inferred that:
 - a. No customer owns a car.
 - b. Only some customers own a car.
 - c. **Every** customer owns a car.
 - d. Most customers own a car.
 - e. None of the above
- 6. A database management system (dbms) is:
 - a. Text editors and compilers
 - b. Data definition, data manipulation language and data model
 - c. Word processor and power point
 - d. **a set of** software for building and maintaining databases.
 - e. none of the above
- 7. The 3 level data architecture for representing data in the dbms are called:
 - a. First, second and third levels
 - b. Foot, middle and top levels
 - c. **External, conceptual** and physical (or internal) levels

- d. All of the above
 - e. none of the above
8. In a database, integrity constraints must not be violated after any database update. Given a database table with schema Client (Id, Fname, Lname, phone), which has a state having the two tuples <1, 'Mark', 'Adam', '416-555-2233'> and <2, 'Pat', 'Moore', '519-333-6161'>, what integrity constraint is violated by inserting the following new tuple into Client <2, 'Peter', 'Hardy', '226-222-1234'>?
- a. Entity Integrity constraint
 - b. Foreign key referential integrity constraint
 - c. Primary Key integrity constraint
 - d. All of the above
 - e. None of the above
9. Some of the main advantages of using a database management system as opposed to the traditional file system of running programs with data in a file are:
- a. Using programming languages such as JAVA to create database applications
 - b. Separation of data from program using the three data level architecture
 - c. Replication of data and programs
 - d. Use of the internet
 - e. none of the above
10. In a database, the 3-level data architecture allows us to define the simple database “to track cars owned by customers” using the dbms data model at the conceptual level. An external level query view of this database is:
- a. Print all courses taken by students
 - b. Print all books borrowed by students
 - c. Print all cars owned by a customer
 - d. All of the above
 - e. None of the above

Section B (35 marks):
This section has 3 questions:

1. **(15 marks)** Given the ER diagram of a UNIVERSITY database of Figure B.1, answer the following questions with this diagram.

Figure B.1: ER diagram for a UNIVERSITY database



Write the relational database **schema** (not the ER schema) for this UNIVERSITY database of Figure B.1, for only the 6 entities and relationships
(Student, Major_dept, Department, Dept_Course, Course, Course_section) including
 (i) all the 3 entities with attributes to save writing time, (ii) all the 3 relationships with attributes,
 (iii) one referential integrity (foreign key) constraint using the relationship ‘**Major_dept**’; your
 answer should indicate both the referencing attribute and the referenced attribute, (iv) explain
 the cardinality constraint (from left to right and from right to left) for the relationship
 ‘**Major_dept**’ entities (e.g., indicating how each record in an entity relates to another entity (ie,
 explain the 1:N or similar that applies?)), (v) explain the participation constraint (total or partial
 and what that means) for both left and right sides of the relationship ‘**Major_dept**’.

Solution:

University Database Component	University SCHEMA for the Database Components
i. 3 Entities in the list of 6 With their attributes (5 marks)	Student (sname, <u>Snum</u> , SSN, phone) Department (dnum, <u>dcode</u> , officenum, phone) Course (cname, title, <u>cnum</u> , credithr) Section (instructor, Semester, year, <u>sectnum</u>) Grade (<u>lettergrade</u>, <u>numgrade</u>) **Detailed scheme: 3 tables expected. Duck (-1.5) for each table missing. There are 3 entities tables. Take -0.5 for each incorrect attribute set or unidentified PK. Deduct up until -5 total.
ii. 3 Relation-ships with their attributes in the list (5 marks)	Major_dept (<u>Snum</u> , <u>dcode</u>) Dept_course (<u>cnum</u> , <u>dcode</u>) Course_section (<u>sectnum</u> , <u>cnum</u>) Student_section (<u>numgrade</u>, <u>Snum</u>, <u>sectnum</u>) **Detailed scheme: at least 4 tables expected. Duck (-1.5) for each table missing. There are 3 relationship tables. Take -0.5 for each incorrect attribute set. Deduct up until -5 total. GA: Note that Major_dept(<u>Snum</u> , <u>dnum</u>) is also correct and Dept_course(<u>cnum</u> , <u>dnum</u>) is also correct.
iii. Referential Integrity constraints for ‘Major_dept’ (from left to right and from right to left entities) (1.5 marks)	iii. The schema for the relationship ‘Major_dept’ is Major_dept (<u>Snum</u> , <u>dcode</u>). This means that the foreign key attribute value for <u>Snum</u> in the Major_dept relationship relation must reference an existing <u>Snum</u> value in the STUDENT entity relation. Also, the foreign key attribute <u>dcode</u> value in the same ‘Major_dept’ relationship relation must reference an existing <u>MName</u> in the DEPARTMENT entity relation.

iv. Full (from left to right and from right to left) Cardinality constraints for the one relationship types: Major_dept, (1.5 Marks)	iv. Each STUDENT record can Major in only one DEPARTMENT (left to right). However, each DEPARTMENT record can have many STUDENTs record majoring in it (right to left).
v. Participation constraints for the relationship 'Major_dept' (for both left to right and right to left). (2 marks)	v. Every STUDENT must be major (total participation) in a DEPARTMENT but not every DEPARTMENT needs to have a majoring in it to exist (partial participation).

****Detailed scheme: 1.5 for (iii) and 1.5 for (iv) and 2 for v. Duck (-1) for any part of the full (left/right) answer in (iii) to (v) that is missing. Any mismatch in the attribute set is -0,5 it is -0.75 when only a left to right or right to left constraint explanation is provided.**

2. **(10 marks)** Given the following four relations for a bank database application:

BANK(Code, Name, Addr)
ACCOUNT(Acct_no, Balance, Type)
CUSTOMER(Ssn, CName, Addr, Phone)
HAS_ACCT(Acct_no, Ssn, Code)

Here, Balance refers to balance in the account; Type refers to the type of account (e.g., savings); Code refers to the bank code. All the other attributes are self-explanatory.

- a. Specify all the foreign keys for this schema. State each foreign key using the following format "foreign key is the attribute ----- of relation ----- that references relation -----".

(5 marks)

- b. State two English queries involving more than one table that can be posed on this database indicating the tables to visit to answer the queries each time.

(5 marks)

Question	Foreign Key
2a (5 marks)	(Note: if you find more foreign keys, insert more lines) i. foreign key is the attribute -----Acct_no----- of relation -- HAS_ACCT --- that references relation --- ACCOUNT - ii. foreign key is the attribute --- <u>Ssn</u> --- of relation -- HAS_ACCT --- that references relation -- CUSTOMER ---

	iii. foreign key is the attribute <u>Code</u> of relation HAS_ACCT that references relation BANK **Detailed scheme: Duck (-1.5) for each of the main foreign keys missing.
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Question	Query in English only	Tables needed to answer query
2b (5 marks)	i. Get the total balance maintained by each customer name and in what bank name Select Customer.Cname, sum(Account.Balance) From Customer, Bank, Account, Has_Account Where Customer.Ssn = Has_Account.SSn and Has_Account.Code = Bank.Code and Has_Account.Acct_no = Account.Acct_no Group by Custmer.Cname; (ii) Print all account types with account number maintained by customer "Smith". Select Account.Type, Acct.Acct_no From Customer, Account, Has_Account Where Customer.Ssn = Has_Account.SSn and Has_Account.Acct_no = Account.Acct_no And Customer.Cname LIKE '%Smith'; **Detailed scheme: Duck (-1.5) for each of the English queries not given. Duck -1 when the query given does not involve two tables. Note that we could have simply given zero for not posing queries using two tables as asked. -1 for not identifying the tables involved in answering the queries and -0.5 for each missing table up to 2.	i. Customer, Bank, Account, Has_Account ii. Customer, Account, Has_Account

3. (10 marks)

a. Design a simple RETAIL store records database with 3 to 4 related tables to track customer purchases of store products and answer the following questions about your database. Provide the database schema in this part with primary keys underlined.

(5 marks)

b. Create a valid instance (a state) of your database with at least 2 records in each table.

(5 marks)

Question	Answers
a. (5 marks)	CUSTOMER(<u>cid</u>, cname, cphone, number_of_goods) PRODUCTS(<u>pid</u>, pname, price) BUYS(<u>cid</u>, <u>pid</u>, cost) **Detailed scheme: Duck (-1.5) for each of 3 main tables not identified correctly with its attributes and primary key (PK) (-0.5 from the -1.5 if only the PK is not identified). Note that the tables must be linked through foreign keys attributes. Total number of markds deducted must not exceed 5.
b. (5 marks)	CUSTOMER ----- <u>cid</u> cname cphone number_of_goods 1 Mary 5199999991 3 2 Peter 5199999992 1 PRODUCTS ----- <u>pid</u> pname price 31 Sugar 2.30 42 Rice 8.00 45 Bread 1.89 50 Broccoli 2.10 BUYS ----- <u>cid</u> <u>pid</u> cost 1 31 2.30 1 42 8.00 1 50 2.10 2 42 8.00 **Detailed scheme: It is -1 for each table not filled. -1 for violating the PK and FK constraints. Basically, if it is a database, the tables should be related through PK and FK attributes and can answer a query using two tables such as "Give me the costs of products purchased by customer 'Mary'. Thus, building unconnected tables or giving random unrelated values with no ICs maintained cause loss of marks.